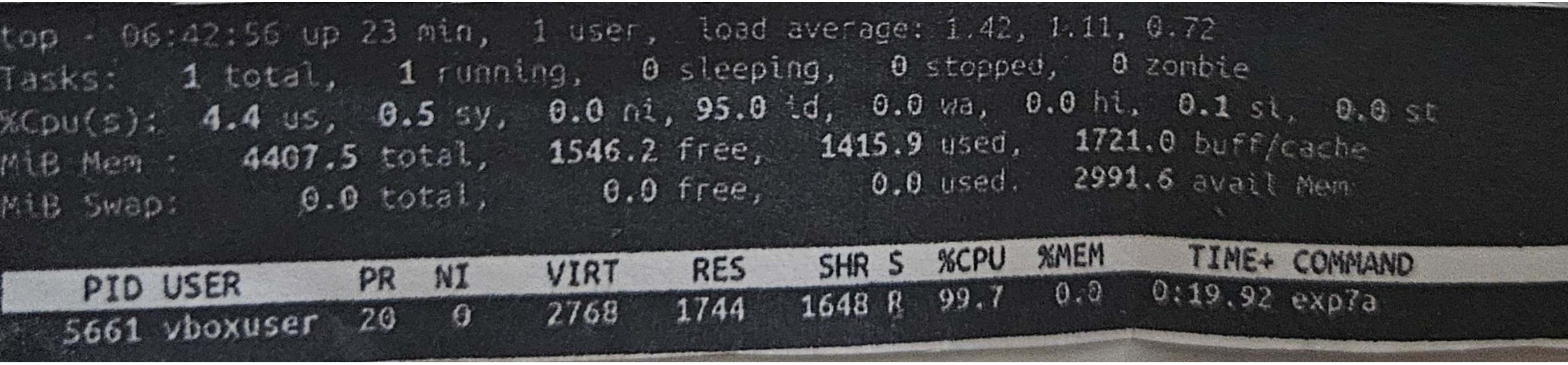

vboxuser@danish:~$ nano exp7a.c
vboxuser@danish:~$ gcc exp7a.c -o exp7a
vboxuser@danish:~$ chmod +x exp7a
vboxuser@danish:~$ ./exp7a

```
  GNU nano 8.7.1
#include<stdio.h>

int main() {
    long long counter = 0;

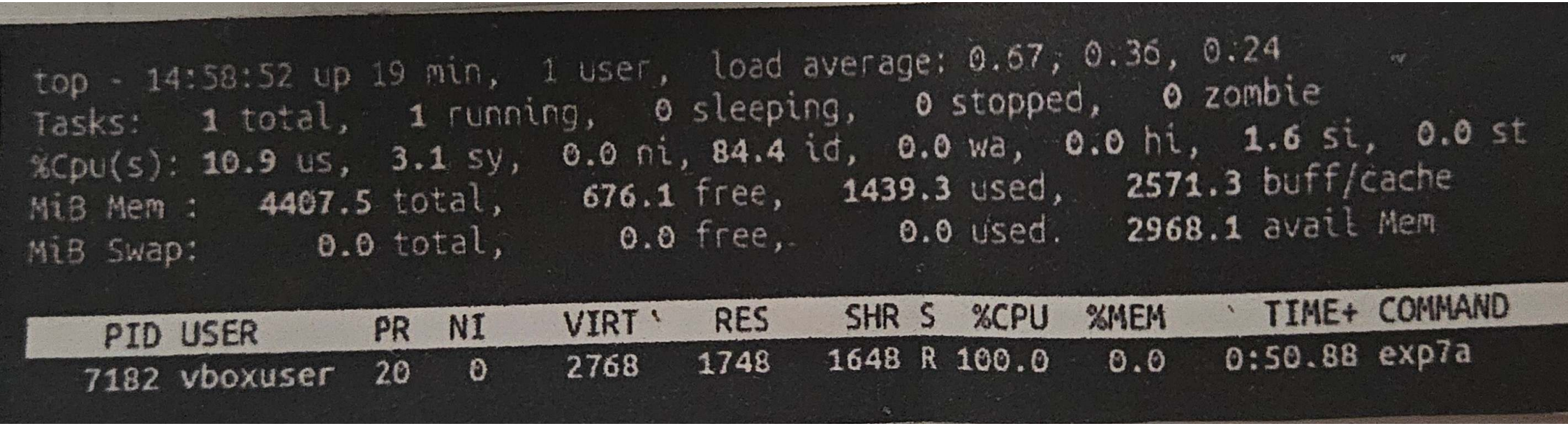
    while (1) {
        counter++;
        if(counter % 100000000 == 0) {
            printf("CPU task running: %lld\n", counter);
        }
    }
    return 0;
}
```

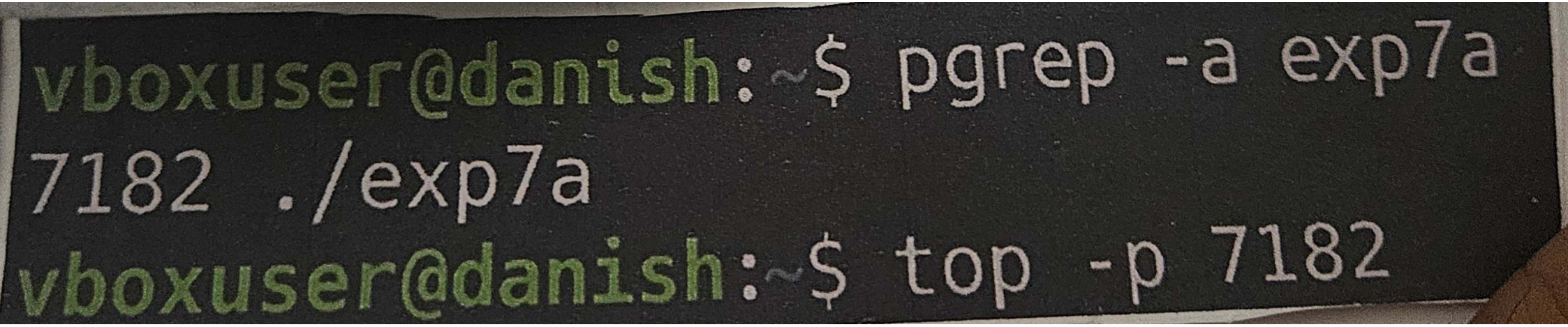

top - 06:42:56 up 23 min, 1 user, load average: 1.42, 1.11, 0.72
Tasks: 1 total, 1 running, 0 sleeping, 0 stopped, 0 zombie
%Cpu(s): 4.4 us, 0.5 sy, 0.0 ni, 95.0 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 4407.5 total, 1546.2 free, 1415.9 used, 1721.0 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 2991.6 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
5661 vboxuser 20 0 2768 1744 1648 R 99.7 0.0 0:19.92 exp?a

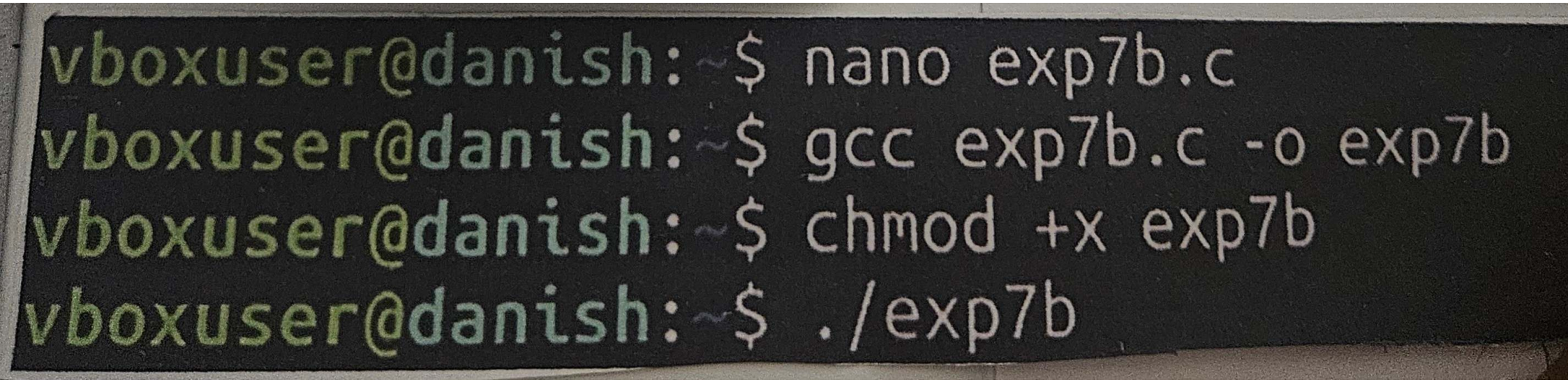
GNU nano 8.7.1

```c
#include<stdio.h>
#include<unistd.h>
int main() {
    FILE *fp;
    printf("Starting I/O task...\n");
    while (1) {
        fp = fopen("dummy.txt", "w");
        if (fp != NULL) {
            fprintf(fp, "Writing disk data\n");
            fclose(fp);
            printf("File operation done\n");
        }
        sleep(1);
        }
        return 0;
}
```

```
vboxuser@danish:~$ pgrep -a exp7b
7424 ./exp7b
vboxuser@danish:~$ top -p 7424
```


top - 14:58:52 up 19 min, 1 user, load average: 0.67, 0.36, 0.24
Tasks: 1 total, 1 running, 0 sleeping, 0 stopped, 0 zombie
%Cpu(s): 10.9 us, 3.1 sy, 0.0 ni, 84.4 id, 0.0 wa, 0.0 hi, 1.6 si, 0.0 st
MiB Mem : 4407.5 total, 676.1 free, 1439.3 used, 2571.3 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 2968.1 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
7182 vboxuser 20 0 2768 1748 1648 R 100.0 0.0 0:50.88 exp7a


vboxuser@danish:~$ pgrep -a exp7a
7182 ./exp7a
vboxuser@danish:~$ top -p 7182


vboxuser@danish:~$ nano exp7b.c
vboxuser@danish:~$ gcc exp7b.c -o exp7b
vboxuser@danish:~$ chmod +x exp7b
vboxuser@danish:~$ ./exp7b